Computers & Industrial Engineering 176 (2023) 108996

Contents lists available at ScienceDirect

Computers & Industrial Engineering

journal homepage: www.elsevier.com/locate/caie

A combined multi-objective multi criteria approach for blockchain-based synchromodal transportation

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ARTICLE INFO

Keywords:
Blockchain
Synchromodal transportation
Supply chain
Multi-objective
Optimization
Linear programming
Decision making
Sustainability

ABSTRACT

Despite our planet's serious environmental challenges, road transportation remains the most important mode of freight shipment, with the broadest coverage and the greatest influence on global economic development. Intermodal transportation may be a more environmentally friendly alternative to heavy road use. However, such a solution is still limited in terms of scalability, service quality, and interoperability. In this context, synchromodal transportation combined with block chain technology may play a critical role in achieving transportation system efficiency and sustainability. In this paper, we study the use of blockchain technology for synchromodal transportation. We build a multi-objective mathematical model for the optimization of transportation cost, transportation time, CO_2 emissions and the blockchain cost. Then, we use several variants of the well-known weighting method and epsilon constraints method to get compromise solutions. We rank the obtained compromise solutions using a variety of multi-criteria decision methods. Then a metaranking method is used to obtain an aggregated ranking. Results show that the proposed methodology may help to ensure the security, the efficiency and the sustainability of synchromodal transportation.

1. Introduction

A revolution in transportation has occurred since the advent of containers in the 1960s. Container volumes have expanded dramatically in recent decades because of rapid globalization and urbanization. The difficulties of handling large volumes of containers have spread from deep-sea terminals to the hinterland network, resulting in substantial use of road transportation, long wait times, and excessive carbon emissions. Maritime transportation, and thus the seaport, continues to be the most important mode of transportation. This mode transports approximately 80% of the total freight volume. For a variety of reasons, seaports nowadays invest not only in infrastructure and equipment to retain their competitiveness in the market, but also in the expansion of their connections to the hinterland network. In the face of inland terminals, the capacity of ports to build reliable and flexible linkages with their hinterland partners adds value to their competitiveness. In fact, the hinterland network, as part of the global supply chain, is responsible for transporting cargo in the most efficient and cost-effective manner possible. In terms of port appeal to merchants and carriers, cheaper and faster hinterland links are, for a seaport, the cornerstone to be more competitive. Also, putting money into reliable backhaul connections could lead to less congestion at terminals and faster container release.

In many countries, the road mode accounts for more than 75% of total inland freight transportation (Marzano et al., 2022). Owing to their flexibility, trucks remain the most efficient solution for shipment of goods in many situations. We refer to the transportation of small parcels and/or goods originated or destinated from inaccessible zones. However, the main drawback for large volume of goods, transported by road, is the huge CO_2 emissions generated. In addition, it leads to congestion, accidents and accordingly to several profit losses, especially in urban areas.

Nowadays, there is a growing awareness of environmental issues related to road transportation. In fact, more influential and better-informed consumer is emerging. More and more consumers request eco-friendly products and services. In addition, Logistics Service Providers (LSP) are considering more efficient transportation modes to reduce environmental impact and costs (El Raoui et al., 2020). Intermodal transportation has the potential to be less expensive and more sustainable than unimodal road transportation (Oudani, 2021). This is due to trains, barges, and ships having lower costs and emissions per ton/km than trucks.

The primary goal of synchromodal logistics is to improve service quality and sustainability while lowering costs by strengthening vertical and horizontal cooperation among stakeholders. Specifically, the service quality is achieved by giving all parties relevant real-time information and by utilizing the most flexible door-to-door shipments. Furthermore, these conditions allow for a more efficient synchronization of operations. In addition, to implement procedures based on the

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https://doi.org/10.1016/j.cie.2023.108996

Available online 16 January 2023
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just-in-time and cross-docking paradigms, adequate synchronization is essential. To avoid storage expenses and unfulfilled demand, the synchromodal transportation operator requests that each cargo arrive just when it is required. Companies, on the other hand, strive to expand and improve their communication networks, as well as optimize their supply chain processes. This is due to the growing demand for information flows such as cash and payment flows, product location and status, and data for real-time controversies. In this regard, Blockchain has emerged as one of the most promising technologies, in recent years, for dealing with the progressive networking of customers, products, and value chains. In fact, Blockchain is a cutting-edge technology that has the potential to significantly improve the transparency, verifiability, and efficiency of supply chains.

The remainder of this paper is arranged as follows. After the introduction, Section 2 provides the preliminary background of intermodal and synchromodal transportation as well as the use of blockchain technology in supply chain. Section 3 presents a review of the existing literature related to synchromodal transportation, blockchain and green supply chain. The multi-objective blockchain-based for synchromodal transportation is presented in Section 4. We provide a problem description, then the model formulation as well as the numerical results using epsilon-constraint method and weighting method. Section 5 is dedicated to multi-criteria decision methods to rank the compromise solutions. Results of three different Multi Criteria Decision Making (MCDM) methods are reported and a metaranking is presented. Section 6 illustrates the findings, discussion of results and eventual implications from managerial and theoretical standpoint. Finally, some concluding remarks along with research limitations are presented in Section 7.

2. Background

2.1. From intermodal to synchromodal transportation

In recent years, with growth of global trade, intermodal transportation is more frequently used as a result of its favorable effects on the economy and the environment (Zhang et al., 2022a). Various management concepts have emerged in the logistics industry over the last few decades as a result of economic factors and environmental concerns (Guo et al., 2022). Modern organizations of transportation include multimodal, intermodal, co-modal, and synchromodal modes. The term “multimodality” refers to the use of several modes, while “intermodality” is a more specific term emphasizing the use of standardized loading units, such as containers and swap bodies, known as Intermodal Transportation Units (ITU). Unimodal road transport still accounts for a large portion of long-haul continental freight flows (Ibnoulouafi et al., 2022). More efficient transport modalities are being examined by logistics service providers in an effort to reduce environmental effect and expenses. Intermodal transportation has the potential to be less expensive and more environmentally friendly than unimodal road transportation. In addition, trucks, typically perform pre and post haulage in intermodal transport. As a first condition to be competitive, the cost and environmental benefits of modal switching, in a intermodal chain, should compensate for the additional transshipment operations, making it less appealing for short distances.

When the unit cost of the consolidation mode is lower than the unit cost of the road mode and compensates for the additional fees of transshipment operations, intermodal transportation is a competitive alternative to door-to-door road transport. Intermodal transportation is extremely successful, if not critical, when the following conditions are satisfied: if the goods are not time-critical; and (ii) if there are a large volume of goods. In fact, despite the high speed of the consolidation mode (for example, rail mode), the overall journey of an intermodal transport chain takes longer than a door-to-door road trip. As a result, for time-sensitive deliveries, intermodal transportation is ineffective. Furthermore, a large quantity of goods is required to benefit from the economy of scale generated by the consolidation mode.

Synchromodality, as an extension of intermodality and co-modality, adds (real-time) flexibility in planning when disturbances occur. Through the flexibility of the service and according to demand and the availability of resources, the decision-maker can change the routes and timings of the vehicles (Zhang et al., 2022b). The main challenge for synchromodal transportation networks is synchronization and the coordination of various modes in the network for the continuous flow of goods (Zahid et al., 2022). The goal of synchromodality is to build a time-sensitive transportation chain that is efficient, environmentally friendly, transparent, and integrated. Collaboration among stakeholders, information technologies, and integrated planning at various decision levels are required for the implementation of synchromodal transportation (Guo et al., 2022). A specific feature in synchromodal transportation is that some input data are identified as the plan is carried out. One of the most common dynamic events is the receipt of new cargo requests, though other dynamics such as demand and journey times are also possible.

2.2. Blockchain for transportation and logistics

For many years, blockchain was a technology that a few insiders knew about, with a concentration on technical dimensions and qualities. However, since Satoshi Nakamoto released Bitcoin in 2009, blockchain has grown in popularity due to its utility in cryptocurrency development. The key advantage of Blockchain technology is that it provides a comprehensive solution for maintaining supply chain security, integrity, data immutability, and traceability. Currently, blockchain is hailed as a solution for improving supply chains and insurance contracts, to name a few hot topics. The blockchain is a distributed ledger that relies on peer-to-peer consensus to validate and authenticate all the stored data (Niranjanamurthy et al., 2019). Specifically, the blockchain is a “chain of blocks” that keeps track of transaction details for a certain period and cannot be easily altered (Oudani et al., 2023). According to openness and data access, the blockchain can be classified into two types: public and private. In terms of ledger access, the public and private networks differ. In a public network, each user has their own copy of the ledger and participates in transaction confirmation independently, whereas in a private network, users must have permission to hold a copy of the general ledger and participate in transaction confirmation. The Fig. 1 illustrates a blockchain system used for a global supply chain.

To comprehend the blockchain system in its broadest sense, it is necessary to employ and investigate several blockchain states:

- Transaction preparation: Partner A begins this stage by creating a transaction that includes information such as the receiver’s public address, a digital signature, and transaction message content.
- Transaction verification: Each node (the machine running the blockchain client software) receives this transaction and verifies the digital signature with party A’s public key. When the verification is successful, the authenticated transaction is recorded in the ledger and waits until all nodes successfully verify the same transaction.
- Block generation: The queued transactions are arranged together, and a block is created by one of the network’s nodes. Bitcoins are rewarded in the Bitcoin blockchain when a Bitcoin node, also known as a miner, creates a block by solving a mathematically complex problem.
- When a block generation is successful, other nodes in the network are processed for an iterative validation process in which the majority of the nodes must reach consensus. There are several methods for reaching agreement: Proof of Stack (PoS), Proof of Work (PoW), Delegated Proof of Stack (DPoS), and Practical Byzantine Fault Tolerance (PBFT). For example, Bitcoin uses PoW

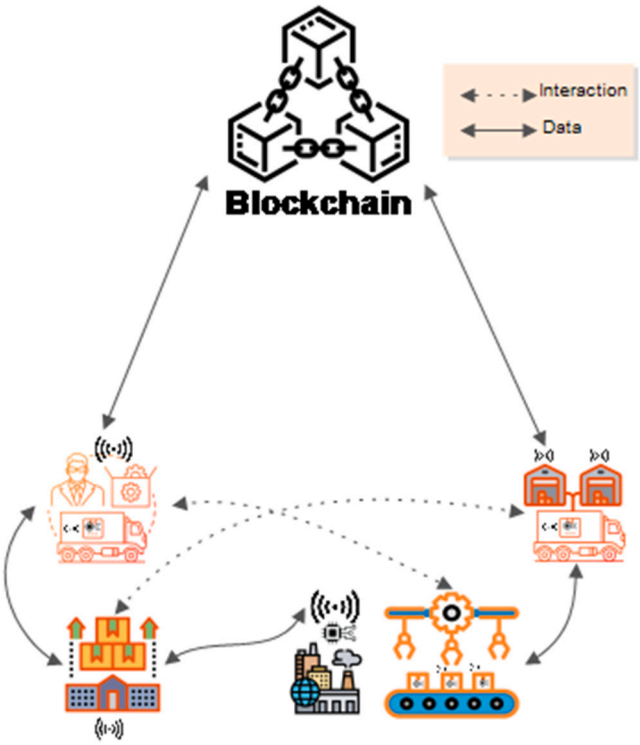


Fig. 1. Blockchain-based supply chain system.

to reach consensus, whereas Ethereum uses PoS. When the consensus mechanism is reached, the blocks are checked and added to the blockchain.

There are many opportunities for supply chain and transportation regarding blockchain technology (Jain et al., 2022; Zkik et al., 2022):

- Supply chain transactions are recorded to ensure information quality, tracking, traceability, and transparency.
- It will help with mode selection as well as the ability to link containers, products, materials, and geographic data to transportation modes in multimodal transportation.
- Smart contracts can make transactions go more smoothly.
- Reduce the amount of time spent between process phases
- Information exchange while protecting proprietary and private data
- The capacity to provide more visible and reliable information on items, processes, etc. can improved circular economy principles.

3. Literature review

The topic covered by the current paper has attracted the attention of researchers and professionals. This is evidenced by the steadily increasing number of papers published regarding the subject.

Many studies have shown that using Blockchain Technology (BT) for supply chain management provides manufacturers with substantial benefits in terms of oversight, information transparency, and traceability (Azzi et al., 2019; Chang & Chen, 2020). It does, in fact, provide a safe and open information resource for the entire supply chain.Furthermore, blockchain makes it simple for various stakeholders to access information and track transactions. In addition, Farooque et al. (2020) explored the possibility of improving energy source workload scheduling to reduce the energy appeal of globally distributed blockchain-based data centers. Also, some studied focused on practical aspects of blockchain in the supply chain. Case studies include cotton industry (Agrawal et al., 2021), chemical industry (Takhar & Liyanage, 2021), and agri-food industry (Teng et al., 2021). Furthermore, Kamble et al. (2021) investigated the advantages and disadvantages of blockchain adoption in the automotive industry. According to their research, blockchain technology provides integration, transparency, and visibility throughout the entire supply chain, which optimizes information flow and reduces costs.

The authors of Ambra et al. (2019c) investigate the state-of-the-art synchromodal and Physical Internet models, as well as the designs, methodologies, and findings proposed in the scientific literature. The main goal is to assess and investigate the relationships between these two concepts in order to understand how they can reinforce one another. The authors of Behdani et al. (2014) developed a mathematical model for designing service schedules for synchromodal freight transport systems. The study (Guo et al., 2020) explored the dynamic shipment matching problem in which a centralized platform matches shipment requests with transport services online. To handle newly received shipping requests, the authors offer a rolling horizon method and construct a heuristic algorithm to generate timely solutions at each decision epoch. A decision-making system for perishable commodities logistics service providers is presented in Lin et al. (2016) to prevent freshness loss via synchromodal transport. Critical success factors of synchromodality and enabling technologies are discussed in Giusti et al. (2019) and Pfoser et al. (2016). The authors of Li et al. (2017) studied the cooperative synchromodal freight transportation planning with several intermodal freight transport providers in various and interconnected service networks. In Van Riessen et al. (2021), a fast algorithm is proposed for a single intermodal corridor network and proofs for the optimality of the result are provided. This study conducted in Ambra et al. (2019b) presented a computational approach for calculating agent movements in a spatially referenced space. For each mode, the model captures stochastic parallel processes. An analytical tool for assessment of modal alternatives is given in Kapetanis et al. (2016) and Port of Piraeus is considered as a case study. Authors in Giusti et al. (2021) discussed relationship between slow steaming and synchromodality and introduced smart steaming. Besides, digital twins are used in Ambra et al. (2019a) for reducing uncertainty in Synchromodal transportation. According to recent studies (Tsertou et al., 2016), cloud computing is a facilitator for addressing the issue of dynamic and synchromodal container consolidation at the Piraeus Container Terminal, Europe's fastest developing port. A transport operator can collect real-time information about uncertainties from the transportation network thanks to the development of digital platforms and the rise of innovative concepts such as synchromodal transport. Such data can be used by machine learning techniques such as Reinforcement Learning (RL) to learn the pattern of uncertainties. As example, synchromodal transportation was implemented using Reinforcement Learning (RL) by Zhang et al. (2022c) The RL in this study uses knowledge gained from interacting with the environment to inform its decisions. Ferjani et al. (2022) presented a synchromodality-based implementation of the modal shift on the Seine Axis. Then, based on the users' preferences, a simulation–optimization framework is proposed to generate different reliable itineraries and proposed the best solution for synchromodal container transport. Peng et al. (2022) proposed a multi-objective optimization model to reduce the total transportation cost and time of a stochastic multimodal transportation network while taking transport schedules into account. The authors used Monte Carlo simulation to deal with the multimodal transportation network's stochastic transportation time. Then, to find the Pareto solutions, they developed a data-driven approach by combining the data pool composed of historical data and the data mining of the machine learning algorithm. Pamucar et al. (2022) introduced a fuzzy multi-criteria decision making (MCDM) model to focus on selecting the best transportation planning strategies for freight companies toward zero-carbon emission in order to provide a tangible basis for policymaking. The Ordinal Priority Approach (OPA) methodology is used specifically to determine the weighting coefficients of the criteria/alternatives, which fits the group of subjective models. Sun

Table 1
Related Works.

Reference	Modeling approach	Objective function	Solving approach	Specific feature? Case study?
Mes and Iacob (2016)	Multi objective k-shortest path	cost, delays, CO_2 emissions	Planning algorithm	time windows corridor from the Netherlands to Italy
Yee et al. (2021)	Markov decision process	–	Numerical case study	comparison synchromdal vs. static intermodal
Resat and Turkay (2019)	multi objective mixed integer program	cost, travel time, CO_2 emissions	epsilon constraint	Traffic congestion Marmara in Turkey
Batarlienė and Šakalys (2021)	stochastic integer program	waiting time journey time	–	container distribution Vilnius and Klaipeda Seaport
Xu et al. (2015)	stochastic integer programming model	expected transportation profit	Integrated Simulated Annealingand Genetic Algorithm	Container allocation problem
Rivera and Mes (2017)	MILP model	time	Matheuristic	sceduling drayage operations in Synchromodal transportation
Pérez Rivera and Mes (2016)	Markov Decision process model	–	Approximate Dynamic programming	planing problem of selecting services and transfers with multi period horizon
Zhang et al. (2021)	multi objective optimization model	cost, time, emissions	ALNS	Pick up and delivery case study in Rhine alpin corridor
Zhang et al. (2022a)	multi objective optimization model	cost, time, emissions	ALNS	Pick up and delivery case study in Rhine alpin corridor

et al. (2022) proposed a multi-objective optimization model for the road–rail intermodal routing problem, with the goal of minimizing the routes' total costs and carbon dioxide emissions. To improve the overall timeliness of the transportation process, pickup and delivery services are simultaneously improved using fuzzy soft time windows to measure their service levels.

There are few works that used optimization and, in particular, linear programming for synchromodal transportation. Delbart et al. (2021) stressed that the most studied types of uncertainty among all planning levels are stochastic demand and stochastic transit times. Some works model the uncertainty and dynamics of synchromodal transport using Markov processes. Linear programming models are used to optimize objective functions such as delivery time, terminal wait time, CO_2 emissions, transportation cost, and so on.

In the Table 1, we provide a synthesis of the most relevant works utilizing optimization in the context of synchromodal transport.

From Table 1, and up to our knowledge, no study from explored literature used blockchain for synchromodal transportation. The contributions of the current paper to fill in this gap are the following:

- We build a multi-objective mathematical model to consider blockchain cost for synchromodal transportation problem
- We use a combined multi-objective and multi-criteria decision making approach for ranking solutions
- We discuss the implications of the developed work

4. A multi-period synchromodal transportation model

4.1. Problem statement

We assume that a set of customers will send goods packed in containers at different time periods (days for examples). The transportation network is composed by different transportation modes. These modes differ by their transportation time, transportation costs and by their CO_2 emissions. To track a container and to record its data along the intermodal chain, blockchain technology may be used in some or in all transportation components (modal links, terminals, etc.).

Table 2
Model parameter notations.

Model Parameters	
I	Set of all customer nodes
M	Set of all transportation modes
P	Set of all time periods
c_{ij}^m	transportation cost between i and j using transportation mode m
e_{ij}^m	the CO_2 emissions generated between i and j using transportation mode m
τ_{ij}^m	the maximum transportation time between i and j using transportation mode m
t_{ij}^m	travel time between i and j using transportation mode m
b_{ij}^m	the fixed cost for using blockchain technology between node i and j using transportation mode m
ct_{ml}	transfer cost between mode m to mode l
t_{ml}	transfer time between mode m to mode l
e_{ml}	the generated CO_2 emissions during transfer between mode m to mode l
w_i^p	the number of containers in customer node i in time period p
q_j	demand of customer j

We take the following assumptions to formulate the problem:

- Modal links have infinite capacity
- A modal link (i, j) using transportation mode m comprise implicitly the use of pre and post haul using road transportation (first and last miles).
- For each link (i, j) using mode m transportation cost, time and emissions are linear function for the number of containers

4.2. Model formulation

To formulate the aforementioned problem, we consider the following parameters (see Table 2):

Let be the following list of decision variables (see Table 3):

The objective functions are as follows:

$$\text{Min } C = \sum_{i\in I, j\in J}\sum_{m\in M}\sum_{p\in P} c_{ij}^m x_{ij}^{mp} + \sum_{m\in M, l\in M}\sum_{i\in I}\sum_{p\in P} ct_{ml} s_{im}^{lp}$$

Table 3
Model decision variable notations.

x_{ij}^{mp}	Binary variable equal to 1 if a container is transported between i and j using transportation mode m in time period p, 0 otherwise.
s_{im}^{lp}	Binary variable equal to 1 if a container is transferred from mode m to mode l in node i in time period p, 0 otherwise.
π_j	time to deliver a container to customer j
y_{ij}^{m}	binary variable equal to 1 if travel time of link (i,j) using mode m is stored in blockchain, 0 otherwise.

$$+ \sum_{i\in I, j\in J} \sum_{m\in M} b_{ij}^{m} y_{ij}^{m} \tag{1}$$

$$\text{Min } T = \sum_{i\in I, j\in J} \sum_{m\in M} \sum_{p\in P} t_{ij}^{m} x_{ij}^{mp} + \sum_{m\in M, l\in M} \sum_{i\in I} \sum_{p\in P} t_{ml} s_{im}^{lp} \tag{2}$$

$$\text{Min } E = \sum_{i\in I, j\in J} \sum_{m\in M} \sum_{p\in P} e_{ij}^{m} x_{ij}^{mp} + \sum_{m\in M, l\in M} \sum_{i\in I} \sum_{p\in P} e_{ml} s_{im}^{lp} \tag{3}$$

The first objective function (1) minimizes the total transportation cost for container shipment and their transfer between modes as well the total blockchain cost. The second objective function (2) minimizes the total travel time for all containers through all transportation modes. This includes time in modal links and time for transfer between modes. The third objective function (3) is to minimize the total CO_2 emissions generated by all transportation modes. These objectives are subjected to the following constraints:

$$\sum_{i\in I} \sum_{m\in M} \sum_{p\in P} x_{ij}^{mp} \geq q_j, \quad \forall j \in I \tag{4}$$

$$\sum_{j\in I} \sum_{m\in M} x_{ij}^{mp} = w_i^p, \quad \forall i \in I, \forall p \in P \tag{5}$$

$$\sum_{m\in M} \sum_{p\in P} x_{ij}^{mp} = \sum_{m\in M} \sum_{p\in P} x_{ji}^{mp}, \quad \forall i \in I, \forall j \in I \tag{6}$$

$$\pi_j \geq \tau_{ij}^{m} y_{ij}^{m} + t_{ij}^{m}(1 - y_{ij}^{m}) \quad \forall i \in I, \forall j \in I, \forall m \in M \tag{7}$$

$$\pi_j \geq 0, \quad y_{ij}^{m}, x_{ij}^{mp}, s_{im}^{lp} \in \{0,1\} \quad \forall i \in I, \forall j \in I, \forall m, l \in M, \forall p \in P \tag{8}$$

Constraints (4) ensure that demand of each customer j is satisfied. Constraints (5) and constraints (6) are conservation flow for the total transported containers in the transportation network. While constraints (7) compute the time to deliver a container to a customer j. When the information, regarding transportation time of a link, is stored in blockchain, the real time transportation time is considered otherwise the maximum transportation time is computed. Constraints (8) are classical restrictions on decision variables.

4.3. Numerical results

The developed model in the previous subsection is a multi-objective mixed integer linear program. A well known method to solve this king of programs is to use the weighting method. In such method the multiple objectives are aggregated to one weighted objective. In doing so, the multi-objective program is converted to single objective model.

To conduct computational experiments, we have randomly generated 15 data-set instances as follows:

We denote by $\alpha = |I|$ the number of nodes (customers) on the transportation network, $\beta = |P|$ the number of time periods and by $\gamma = |M|$ the number of transportation modes. The associated parameters to the instance with α number of customers, β time periods and γ transportation modes are formed as follows: the parameters $c_{ij}^{m}, b_{ij}^{m}, t_{ij}^{m}, e_{ij}^{m}$ are real numbers drawn from the interval $[1, 70]$. The parameters w_i^p and q_j are integer numbers generated from the interval $[10\alpha, 100\alpha]$. The transfer parameters ct_{ml}, t_{ml}, e_{ml} between modes (cost, time and emissions) are randomly drawn from the interval $[1, 20]$. For all generated instances, we take $\gamma = 4$. We adopt the following notation for the name of instances: "Instance.$\alpha.\beta$", where the number of nodes α belongs to the set $\{25, 50, 100, 125\}$ and the number of time periods β is taken from the set $\{5, 10, 15\}$. For instance, "Instance.50.10" stands for an instance with 50 nodes and 10 time periods. All experiments are conducted using a personal computer with the following technical specification: Intel Core i7 processor with 2.6 GHz speed and 16 GB RAM.

The weighting sum with equal weights version is to use equal weights for all the objectives. Specifically, the weights are computed as follows:

$$w_i = \frac{1}{n}$$

Where n is the number of objective functions.

We used also the rank sum method where the weights are computed as follows:

$$w_i = \frac{2(n+1-i)}{n(n+1)}$$

Another weighting method variant is the Rank Order Centroid (ROC) where the weights are computed as follows:

$$w_i = \frac{1}{n} \sum_{k=i}^{n} \frac{1}{k}$$

The multi-objective model was also solved using the epsilon constraint method. It is based on the formulation of an auxiliary model in which the primary objective is optimized and the remaining objectives are treated as auxiliary constraints with epsilon bounds applied to their values.

The results of all these methods are summarized in Table 4.

Using the above-mentioned multi-objective methods, 240 solutions are generated. We conducted an experimental analysis to eliminate dominated solutions. Then, we saved the non-dominated and diverse solutions in a pool (a set of 15 solutions). The objective of the next section is to rank these solutions using MCDM methods.

5. A MCDM approach

In this section, we use several MCDM approaches to rank the pool of solutions given by the multi-objective methods (weighting methods and epsilon constraint method). Then an aggregated ranking is performed.

5.1. MARCOS method

Marcos method is a new method introduced in Stević et al. (2020). MARCOS Method has the following steps:

- **Step 1**
  The MARCOS method starts by using an initial decision-making matrix. This matrix contains evaluation of m alternatives over n criteria by the decision-maker. In most cases, this matrix may be an aggregation of evaluation matrices by a group of experts. Let us denote by P this initial matrix.

  $$P_1 = (p_{ij})$$

- **Step 2** Th initial decision matrix of step 1 is extended by adding two rows representing ideal and anti-ideal solution noted id_j and aid_j respectively for each column. As the criteria may be divided into two groups: the first is the benefit criteria while the second represent the cost criteria, the ideal and anti-ideal solution are computed as follows for each alternative:

  $$id_j = \max_i(p_{ij})$$

  if j is benefit and

  $$id_j = \min_i(p_{ij})$$

  if j is a cost criteria

  $$aid_j = \min_i(p_{ij})$$

Table 4
Results of multi-objective methods.

Instance	Equal sum method				Rank sum method				ROC method				ϵ-constraint			
	f^*	C	T	E	f^*	C	T	E	f^*	C	T	E	f^*	C	T	E
Instance.25.5	455	760	950	1140	1520	1900	3240	3780	1900	3240	3780	4050	1754	4840	3976	2863
Instance.25.10	405	1080	1350	1620	2160	2700	3240	3780	2700	3240	3780	4050	2600	4640	3909	3980
Instance.25.15	93	114	242	174	238	258	328	348	258	328	348	658	258	398	287	876
Instance.50.5	81	130	130	222	247	298	444	464	298	444	464	453	268	894	386	487
Instance.50.10	171.5	399	482	591	798.75	1011	1427	1687	1011	1427	1687	1747.25	1911	1327	1397	1768
Instance.50.15	515	760	1310	1500	2240	2800	3240	3780	2800	3240	3780	4050	2500	3134	2867	3850
Instance.75.5	360	1080	870	1140	1200	1500	3240	3780	1500	3240	3780	4050	1500	3239	7639	3950
Instance.75.10	95	114	182	171	228	248	328	348	248	328	348	658	278	128	846	858
Instance.75.15	101	130	195	195	262	313	444	464	313	444	464	453	363	765	387	553
Instance.100.5	284.9	592.2	729.2	888.3	1184.2	1489.1	2158	2530	1489.1	2158	2530	2568.1	1369.1	2839	2567	2768
Instance.100.10	560	1030	1400	1680	2240	2800	3240	3780	2800	3240	3780	4050	2200	2973	3875	2050
Instance.100.15	300	720	750	900	1200	1500	3240	3780	1500	3240	3780	4050	1100	3029	3865	6050
Instance.125.5	96	178	182	171	228	248	328	348	248	328	348	658	228	289	248	658
Instance.125.10	107	126	194	195	262	313	444	464	313	444	464	453	343	372	874	876
Instance.125.15	352.23	697.62	893.62	1081.62	1442.04	1808.85	2489.42	2912.75	1808.85	2489.42	2912.75	3081.98	1780	2876	2987	3087

if j is a benefit and

$$aid_j = \max_i(p_{ij})$$

if j is a cost
The ideal solution is the alternative with the best characteristic while the anti-ideal solution is the worst alternative. The obtained extended matrix is denoted

$$P_2 = (e_{ij})$$

- **Step 3** In this step a normalization of the extended matrix is made using the following two equations:

$$n_{ij} = \frac{e_{ij}}{id_j}$$

if j is a benefit

$$n_{ij} = \frac{id_j}{e_{ij}}$$

if j is a cost
- **Step 4** compute the weighted matrix $L = (l_{ij})$. The elements of the latter matrix are obtained by the equation:

$$l_{ij} = n_{ij} \times w_j$$

Where w_j is the weight coefficient of criterion j
- **Step 5** compute the utility degree of each alternative by using the equation:

$$U_i^- = \frac{S_i}{S_{aii}}$$

$$U_i^+ = \frac{S_i}{S_{ai}}$$

Where

$$S_i = \sum_{j=1}^{n} l_{ij}$$

- **Step 6** compute the final utility of each alternative by computing the following two terms:

$$O_i^- = \frac{(U_i^+)}{(U_i^+ + U_i^-)}$$

$$O_i^+ = \frac{(U_i^-)}{(U_i^+ + U_i^-)}$$

The final utility of each alternative i is given by:

$$F_i = \frac{(U_i^+ + U_i^-)}{(1 + \frac{(1-O_i^+)}{(O_i^+)} + \frac{(1-O_i^-)}{(O_i^-)})}$$

Table 5
MARCOS Steps 1–5 results.

Solutions	S_i	U_i^-	U_i^+
S_1	0.45354442	4.74239165	0.77901979
S_2	0.26796077	2.80187532	0.46025644
S_3	0.21029259	2.1988801	0.36120406
S_4	0.17812373	1.86251328	0.30594999
S_5	0.13402954	1.40145161	0.23021265
S_6	0.41023687	4.28955532	0.7046336
S_7	0.26796077	2.80187532	0.46025644
S_8	0.16507725	1.72609548	0.28354101
S_9	0.1445477	1.51143251	0.24827891
S_{10}	0.10124716	1.05866958	0.17390477
S_{11}	0.38954831	4.07322982	0.66909839
S_{12}	0.2073742	2.16836458	0.35619136
S_{13}	0.15886898	1.66117997	0.27287752
S_{14}	0.1942771	2.03141757	0.33369544
S_{15}	0.10124716	1.05866958	0.17390477

Table 6
MARCOS Steps 6–7 results.

Solutions	O_i^-	O_i^+	F_i	Final ranking
S_1	0.1410907	0.8589093	0.15430859	8
S_2	0.1410907	0.8589093	0.17036498	6
S_3	0.1410907	0.8589093	0.12555027	15
S_4	0.1410907	0.8589093	0.23976231	3
S_5	0.1410907	0.8589093	0.21785084	4
S_6	0.1410907	0.8589093	0.15159267	9
S_7	0.1410907	0.8589093	0.14719772	11
S_8	0.1410907	0.8589093	0.15007739	10
S_9	0.1410907	0.8589093	0.09199841	1
S_{10}	0.1410907	0.8589093	0.14488735	12
S_{11}	0.1410907	0.8589093	0.17588497	5
S_{12}	0.1410907	0.8589093	0.16672983	7
S_{13}	0.1410907	0.8589093	0.29955687	2
S_{14}	0.1410907	0.8589093	0.13772042	13
S_{15}	0.1410907	0.8589093	0.11643134	14

- **Step 7** the last step is the ranking of alternatives based on their final value of utility. The best alternative has the highest utility and vice-versa.

Results of MARCOS method steps 1–5 are given in Table 5
Results of MARCOS method steps 6–7 are given in Table 6

5.2. VIKOR method

The VIKOR method is a well-known multi-criteria decision-making method (MCDM) for selecting and ranking alternatives. This method assumes that the decision maker (DM) has several options called alternatives. It is used in situations where the DM should determine compromise solutions for complex problems with conflicting criteria.

The VIKOR method is based on a specific measure of closeness to the ideal solution (Opricovic & Tzeng, 2004). Several versions of the original VIKOR method have been developed in the literature. The VIKOR method take a decision matrix $M = (m_{ij})$ as initial input data.

- **Step 1** The first step of this method is to compute the m_j^* and m_j^- best and the worst values respectively for the objective functions from the matrix M.

$$m_j^* = \max_j m_{ij}$$

$$m_j^- = \min_j m_{ij}$$

With $i = 1, \cdot, m$, where m is the solution number, $j = 1, \cdot, n, n$ presents the number of objective function and m_{ij} is the objective function value j taken by the solution i.

- **Step 2** In the second step, we compute the intermediary matrix $I = (I_{ij})$ as follows:

$$I_{ij} = w_j \frac{(m_j^* + m_{ij})}{(m_j^* - m_j^-)}$$

- **Step 3** we compute U_i that represents the maximum utility and R_i the opponent minimum individual regret is calculated. The equations to compute Si and Ri are:

$$U_i = \sum_{j=1}^{n} I_{ij}$$

$$R_i = \max_j (I_{ij})$$

- **Step 4** In this step, we calculate the following parameters:

$$U^* = \min_j U_i$$

$$R^* = \min_j R_j$$

$$U^- = \max_j U_j$$

$$R^- = \max_j R$$

- **Step 5** Finally, the ranking measure F_i is computed using the following formula:

$$F_i = \alpha \frac{(S_i - S^*)}{(S^- - S^*)} + (1 - \alpha) \frac{(R_i - R^*)}{(R^- - R^*)}$$

The parameter $\alpha \in [0, 1]$.

Considering the above steps, Table 7 gives the final ranking of 15 compromise solutions.

5.3. A possibility degree approach

The first step for this approach is to perform a normalization of the initial decision matrix (El Raoui et al., 2022; Raoui et al., 2021). Normalization is required before using an aggregation function so that the decision matrix is invertible and that all its elements are at the same scale. There are a variety of normalizing approaches (El Raoui et al., 2021). An example is to proceed as follows for a decision matrix $X = (x_{ij})$:

- **Step 1:**

$$n_{ij} = \frac{x_{ij}}{\sum_{i=1}^{n} x_{ij}^2} \tag{9}$$

For benefit criterion, and

$$n_{ij} = 1 - \frac{x_{ij}}{\sum_{i=1}^{n} x_{ij}^2} \tag{10}$$

For cost criterion.

Table 7
VIKOR results.

Solutions	U_i	R_i	Q_i	Final ranking
S_1	8,10053042	2,80294594	0,23136223	7
S_2	9,33685418	3,2328909	1	1
S_3	8,04362138	2,82324377	0,23383886	6
S_4	8,644027	3,02806734	0,60336562	2
S_5	8,02285249	2,82324377	0,22774726	9
S_6	8,17512218	2,89280997	0,3381035	3
S_7	8,09030388	2,87638718	0,29771707	4
S_8	8,00548558	2,85996439	0,25733064	5
S_9	7,92066727	2,8435416	0,21694421	8
S_{10}	7,83584897	2,82711881	0,17655778	10
S_{11}	7,75103067	2,81069602	0,13617135	11
S_{12}	7,66621236	2,79427323	0,09578491	12
S_{13}	7,63213831	2,77785044	0,07028197	13
S_{14}	7,64298983	2,76142765	0,05795589	14
S_{15}	7,67923616	2,74500486	0,0530782	15

- **Step 2:** For each solution i, an interval $I_i = [L_i, U_i]$ is computed by solving the two linear programs P_1 and P_2:

$$(P_1)\begin{cases} L_i = \text{Min} \sum_{j=1}^{m} w_j x_{ij} \\ \text{subject to} \\ w_j \geq w_{j+1} \quad \forall j \in \{1, \ldots, m-1\} \\ \sum_{j=1}^{m} w_j = 1 \\ w_j \in [0, 1] \end{cases}$$

$$(P_2)\begin{cases} U_i = \text{Max} \sum_{j=1}^{m} w_j x_{ij} \\ \text{subject to} \\ w_j \geq w_{j+1} \quad \forall j \in \{1, \ldots, m-1\} \\ \sum_{j=1}^{m} w_j = 1 \\ w_j \in [0, 1] \end{cases}$$

- **Step 3:** In this step, a reference interval corresponding to the biggest lower bound is determined $I^* = [L^*, U^*]$, then an interval comparison procedure is performed with regard to the reference interval as follows:
For two intervals $I = [a, b]$ and $J = [c, d]$
  - If $I \cap J = \emptyset$ then

$$P(I \geq J) = \begin{cases} 0 \text{ if } b \leq c \\ 1 \text{ if } a \geq d \end{cases}$$

  - If $I \cap J \neq \emptyset$ then

$$P(I \geq J) = \frac{\int_c^b f(x)dx}{\int_c^b f(x)dx + \int_a^d f(x)dx}$$

- **Step 4:** Finally, for every solution S_i the value $P(I_i \geq I^*)$ is calculated. solutions are then ranked based on the probability degree values that were computed.

Considering the above steps, 15 compromise solutions are ranked using the possibility degree approach. Results are given in the Table 8

5.4. A metaranking approach

One of the primary challenges with multi criteria decision making approaches is that different MCDM methods yield distinct and potentially conflicting rankings, even when they use the same input, which

Table 8
Possibility degree results.

Solutions	L_i	U_i	P_i	Final ranking
S_1	0,00932725	0,99100969	0,50071305	9
S_2	0,0127071	0,98484349	0,5	13
S_3	0,00948681	0,99308165	0,50128286	7
S_4	0,01109696	0,98990985	0,50088578	8
S_5	0,00948681	0,99350739	0,5013914	6
S_6	0,01003368	0,99348895	0,50152691	14
S_7	0,00990458	0,99449512	0,50175014	12
S_8	0,00977548	0,9955013	0,50197312	11
S_9	0,00964637	0,99650748	0,50219583	10
S_{10}	0,00951727	0,99751365	0,50241829	15
S_{11}	0,00938817	0,99851983	0,50264049	5
S_{12}	0,00925907	0,999526	0,50286243	1
S_{13}	0,00912997	0,99946782	0,50281461	2
S_{14}	0,00900087	0,99921575	0,50271766	4
S_{15}	0,00887176	0,99955941	0,50277166	3

Table 9
Metaranking results.

Solutions	VIKOR	Possibility degree	MARCOS	Aggregated ranking
S_1	7	9	8	8
S_2	1	13	6	7
S_3	6	7	15	1
S_4	2	8	3	9
S_5	9	6	4	3
S_6	3	14	9	12
S_7	4	12	11	4
S_8	5	11	10	11
S_9	8	10	1	6
S_{10}	10	15	12	10
S_{11}	11	5	5	5
S_{12}	12	1	7	2
S_{13}	13	2	2	13
S_{14}	14	4	13	14
S_{15}	15	3	14	15

implies that establishing an overall aggregated ranking of options is critical. Another concern is that no method for evaluating the quality of a ranking generated by an MCDM algorithm has been established. We use the proposed methodology in Peng et al. (2011) to design a "fusion ranking" approach. The aforementioned VIKOR, MARCOS and possibility degree approaches are used to build such meta-ranking approach. Through three steps, the proposed fusion approach aims to identify a compromise ranking based on the conflicting MCDM rankings. The first step is to apply the three above-mentioned MCDM methods to rank compromise solutions. We use ranking scores generated by these MCDM methods as input for the next step. In the second step, weights of MCDM techniques are computed. Indeed, a combinatorial evaluation model is used to find the weights (Peng et al., 2011). Let us consider ranking scores produced by each MCDM approach as a vector u_i and let us note by u_0 the following linear combination:

$$u_0 = \sum_{i=1}^{n} w_i u_i$$

Where n is the number of MCDM methods and w_i are decision variables/weight of each vector u_i. The combinatorial evaluation model is given by:

$$\text{Min} \sum_{i=1}^{n} \|u_0 - u_i\| = \sum_{i=1}^{n} \left\| \sum_{i=1}^{n} w_i u_i - u_i \right\|$$

Subject to:

$$\sum_{i=1}^{n} w_i = 1$$

$$0 < w_i \leq 1, \quad \forall i \in \{1, \ldots, n\}$$

In the final step, the weights found by solving the combinatorial evaluation model in the second step are used to generate a second ranking of compromise solutions. We use the rankings scores of MCDM methods obtained in the first step as criteria in the third step. Results of this approach are summarized in Table 9.

6. Implications

6.1. Theoretical implications

Our study contributes to the theory in several directions. First, it advances the debate on the issue of synchromodal transportation and blockchain-based supply chains. Up to date, most of the literature focus on the positive aspect of synchromodal transportation to improve the sustainability and greenness level of supply chains. Our study provides new insights on how to use blockchain-based system to design an efficient synchromodal transportation chain. Second, the study advances operations research within supply chain management field by proposing a novel multi-objective and advanced mathematical models for time, cost and blockchain cost minimization. The models consider the benefits of Blockchain to track containers and to provide the real estimation of travel time. Third, the ranking of the output compromise solutions of the mathematical models using several multi-criteria decision making methods may allow the decision maker the opportunity to choose the most suitable solution from different perspectives. This achievement could constitutes a basis for initiating further works to solve the issue of using blockchain in synchromodal transportation.

6.2. Managerial implications

Aside from the meaningful implications to theory. Our study proposes several implications at managerial side. First, our study proposes and implements a decision support to help decision makers rank solutions from near-optimal alternatives. This enables also supply chain managers to have a detailed dashboard ranking alternatives solutions depending on the context. In addition, such tool may support them in the integration of blockchain technology for other level of supply chain. Second, we proposed a fusion ranking approach to cope with shortcoming of conflicting rankings by MCDM methods, which may assist supply chain practitioners and managers to enhance their understanding when using ranking alternatives on synchromodal transportation. Finally, our study provides a methodology to exploit solution from Pareto front when an industrial problem is formulated as multi-objective model. This is very useful for managers and practitioners especially when decision makers require an accurate ranking of possible alternatives.

7. Conclusion

This study aims to contribute to the growing area of research in synchromodal transportation by exploring the use of blockchain technology for improving its service quality. The current paper provided an important opportunity to advance the understanding of using blockchain technology in synchromodal transportation. We built a multi-objective integer linear program to minimize transportation and blockchain cost, travel and waiting time as well as the CO_2 emissions. We used various variants of the well-known weighting method and epsilon-constraint method to obtain a set of compromise solutions. Then, best compromise solutions are ranked using several multi-criteria decision making methods. To ensure more accurate ranking, we used an aggregated ranking method to get a new ranking. The findings of our work should make an important contribution to the field of using blockchain technology for transportation systems especially synchromodal transportation.

While we provided an efficient modeling and solving methodology for using blockchain technology in synchromodal transportation, a potential opportunity for future research is to consider uncertainty

in the proposed linear programs. Further research might explore and investigate to simultaneously optimize the blockchain cost, the cost of each type of energy as well as its CO_2 footprint. Furthermore, it may be interesting to apply the proposed framework on a practical case study. This should prove the genericity of our methodology. From methodological standpoint, instead of using a linear solver for small and medium instances for the multi-objective developed model, using heuristics approaches, may be explored in a future study to solve larger sized instances.

CRediT authorship contribution statement

Mustapha Oudani: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing – original draft, Writing – review & editing, Visualization.

Data availability

Data will be made available on request.

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